Intermediate Elective

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Set up

Packages

```
# general use
library(tidyverse)
library(janitor)

# file organization
library(here)

# customizing visualizations
library(RColorBrewer)
```

Data

```
# reading in data
veg <- read_csv(here("data", "veg.csv"))

metadata <- read_csv(here("data", "vp_veg_metadata.csv"))
```

1. Explain your inspiration

I chose Cristina Cametti's "IKEA most popular designers" for my inspiration. This is a reasonable choice for the vegetation dataset because there are a large number of different grass species present on the NCOS mesa, just as there are a large number of different IKEA designers. Each grass species found on the NCOS Mesa will be represented by its own color. The height of each bar will represent the total count of unique observations for each species in 2025. The labels at the end of each bar denote count.

2. Planning

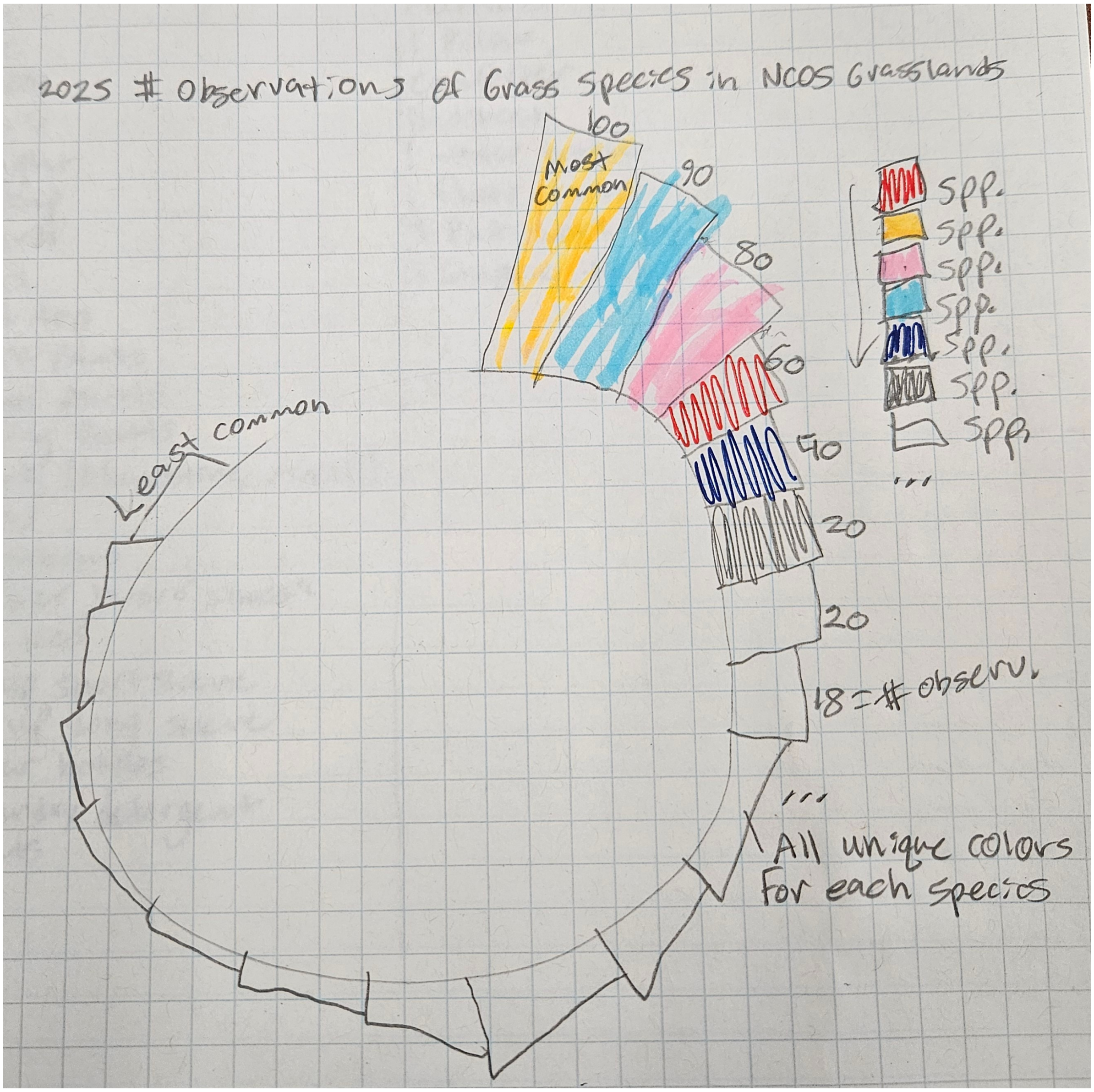


Figure 1: Sketch of planned circular grass species plot

3. Code

Wrangling

```r
# rolling up Hordeum brachyantherum brachyantherum to species level
veg <- veg |>
  mutate(
    psoc = case_when(
      psoc == "Hordeum_brachyantherum_brachy." ~ "Hordeum_brachyantherum",
      TRUE ~ psoc
    )
  )

# create new list value of desired poaceae species
poaceae_species <- c(
  "Bromus_hordeaceus",
  "Festuca_perennis",
  "Stipa_pulchra",
  "Polypogon_monspeliensis",
  "Brachypodium_distachyon",
  "Festuca_myuros",
  "Avena_barbata",
  "Hordeum_marinum",
  "Avena_fatua",
  "Parapholis_incurva",
  "Bromus_diandrus",
  "Distichlis_spicata",
  "Bromus_catharticus",
  "Hordeum_murinum",
  "Cortaderia_selloana",
  "Festuca_bromoides",
  "Cynodon_dactylon",
  "Poa_annua",
  "Festuca_microstachys",
  "Polypogon_viridis",
  "Crypsis_schoenoides",
  "Polypogon_interruptus",
  "Hordeum_brachyantherum",
  "Bromus_madritensis",
  "Elymus_triticoides"
)
```

```r
# create a new object from veg
grassland <- veg |>
  # keep only 2025 data
  filter(
    year == 2025,
    # keep only NCOS Mesa grassland transects
    str_detect(transect_name, "GL"),
    site == "ncos",
    # keep only grass species
    psoc %in% poaceae_species
  )

# creating new object from grassland
grass_counts <- grassland |>
  # calculating number of observations for each species
  count(psoc, sort = TRUE, name = "occurrence_count") |>
  # arrange in descending order
  arrange(desc(occurrence_count))
```

Coding

```r
# create a new object from grass_counts
grass_plot <- grass_counts

# create cleaned species names for the legend
grass_plot <- grass_plot |>
  mutate(
    psoc_clean = str_replace_all(psoc, "_", " ")
  )

# set the number of empty bars to create a visual gap in the circle
empty_bar <- 2

# create empty rows that will become the gap in the circular plot
to_add <- data.frame(
  psoc = rep(NA, empty_bar),
  occurrence_count = rep(NA, empty_bar),
  psoc_clean = rep(NA, empty_bar)
)
```

```r
# add the empty rows to the plot dataset
grass_plot <- bind_rows(grass_plot, to_add)

# create an id column to control where each bar appears around the circle
grass_plot <- grass_plot |>
  mutate(id = row_number())

# calculate maximum observation count for proportional plot sizing
max_count <- max(grass_plot$occurrence_count, na.rm = TRUE)

# create label data for count labels around the circular plot
label_data <- grass_plot |>
  filter(!is.na(occurrence_count)) |>
  mutate(
    angle = 90 - 360 * (id - 0.5) / nrow(grass_plot),
    hjust = ifelse(angle < -90, 1, 0),
    angle = ifelse(angle < -90, angle + 180, angle)
  )

# create one color for each grass species
palette <- colorRampPalette(brewer.pal(11, "Spectral"))(
  length(unique(na.omit(grass_plot$psoc_clean)))
)

# create the ggplot with base layer grass_plot
p <- ggplot(grass_plot) +
  # draw one bar for each grass species with occurence count
  # as the y axis
  geom_col(
    aes(
      x = as.factor(id),
      y = occurrence_count,
      fill = psoc_clean
    ),
    width = 1
  ) +
  # apply custom colors by species
  scale_fill_manual(
    values = palette,
    # label legend
    name = "Grass species",
    # remove NA from legend from added empty rows
```

```
  na.translate = FALSE
) +
# Setting plot proportions
scale_y_continuous(
  limits = c(-max_count * 0.70, max_count * 1.12),
  expand = c(0, 0)
) +
# convert the bar chart into a circular layout
coord_polar(start = 0, clip = "off") +
# add count labels around the outside of the circle
geom_text(
  data = label_data,
  aes(
    x = as.factor(id),
    y = occurrence_count + max_count * 0.06,
    label = occurrence_count,
    hjust = hjust,
    angle = 0,
  ),
  size = 2.4,
  fontface = "bold",
  color = "black",
  inherit.aes = FALSE
) +
# add title inside the center hole
annotate(
  geom = "text",
  x = 0,
  y = -max_count * 0.52,
  hjust = 0.5,
  vjust = 0.65,
  label = "2025 n of\nObservations of\nGrass Species in\nNCOS Grasslands",
  size = 4.3,
  lineheight = 0.85,
  fontface = "bold",
  color = "black"
) +
# add a theme
theme_minimal() +
# remove axes, gridlines, and extra plot clutter
theme(
  axis.text = element_blank(),
```

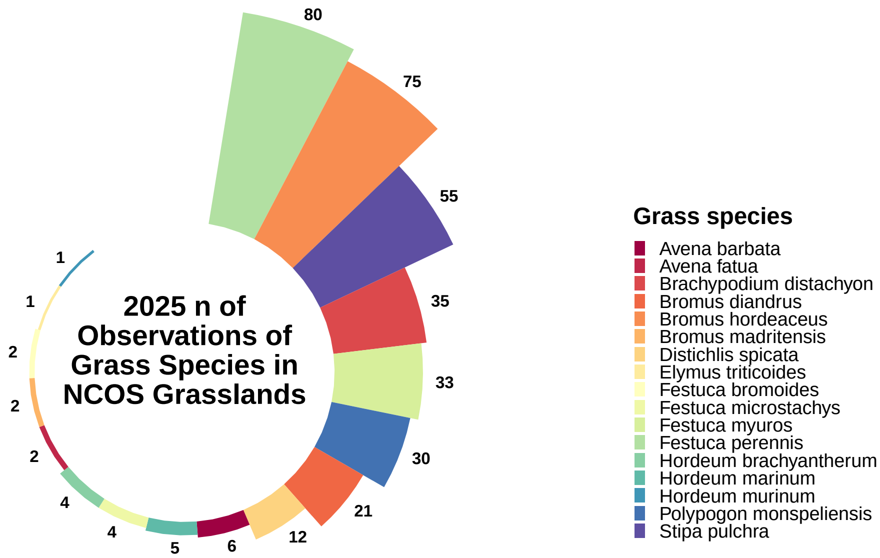
```r
    axis.title = element_blank(),
    panel.grid = element_blank(),

    # remove external title spacing to make room for bigger plot
    plot.title = element_blank(),
    plot.subtitle = element_blank(),

    # reduce outer blank space
    plot.margin = margin(0, 0, 0, 0),

    # legend sizing and positioning
    legend.position = "right",
    legend.direction = "vertical",
    legend.title = element_text(size = 10, face = "bold"),
    legend.text = element_text(size = 8),
    legend.key.size = unit(0.20, "cm"),
    legend.spacing.y = unit(0.01, "cm"),
    legend.box.spacing = unit(0.05, "cm"),
    legend.margin = margin(0, 0, 0, -8),
    legend.box.margin = margin(0, 0, 0, -10),
  ) +
# keep legend in one vertical column
guides(
  fill = guide_legend(
    ncol = 1,
    byrow = TRUE,
    keywidth = unit(0.20, "cm"),
    keyheight = unit(0.20, "cm")
  )
)

# show the finished plot
p
```




```
# save the final circular grass species plot as a PNG file
ggsave(
  filename = here("code", "grass_species_commonness.png"),
  plot = p,
  width = 7,
  height = 6,
  units = "in",
  dpi = 300,
  bg = "white"
)
```

4. Describing my process

My process using someone else's code to create a visualization presented many more challenges than I anticipated. Rather than having the ability to plug my variables into their code as I anticipated, I was forced to change so much of it in order to fit my variables and context that I had to entirely rewrite using the original skeleton for reference.

Most of the challenges I encountered had to do with tedious corrections of figure spacing. Adding more values to the circular plot changed the proportions. The proportions were also heavily skewed by the largest value, which in my case is significantly larger than the smallest value (80 and 1 respectively).

My visualization is different from any that I have made for class or assignments in that it utilizes a unique circular geometry in which the length of the bar rather than height represents its value along the y axis. Further, I was free with this creative visualization to play around with structure, such as placing the title in the center of the plot.